Type MS · Read or estimate a value from a graph, chart, table or image

Answer 1 of 11 · Question 6 part b.i · 1 mark

2025 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 6 | (a) | (i) | any two from:<br>• alcohol absorbed/passes into blood;<br>• rapidly/immediately/straight away/steep increase;<br>• by stomach; | ignore intestines | 2 |
| | | (ii) | • (alcohol carried to) liver;<br>• broken down/detoxified;<br>• level (in blood) falls; | | 3 |
| | (b) | (i) | • any time after (including) 2.4 hours/2 hours 24 mins; | allow a time of between 1-20 minutes | 1 |
| | | (ii) | • alcohol is a depressant;<br>• slows reaction/increases reaction time;<br>• drivers can't respond as quickly/less alert;<br>• leads to accidents; | | 4 |
| | (c) | | • collecting duct less permeable (in absence of ADH);<br>• less water reabsorbed;<br>• into blood;<br>• volume of urine (therefore) increased/more urine produced;<br>• and is more dilute/less concentrated; | accept ref fewer aquaporins/water channels | 5 |
| | | | | | 15 marks |

## Type MS · Read or estimate a value from a graph, chart, table or image

### Answer 1 of 11 · Question 6 part b.i · 1 mark

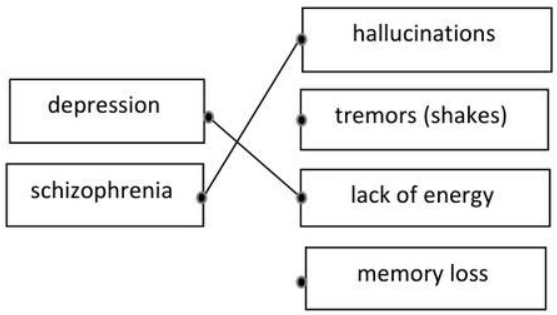
2025 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 2 of 11 · Question 2 part a.i · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) | 2002; | | 1 |
| (ii) | 2005 (1.4) | | 1 |
| (iii) | 2.75:2.6;<br>1.1/1.06 :1; | | 2 |
| (iv) | low income/low self esteem/(domestic) violence/low social status/substance abuse; | | 1 |
| (b) |  | | 2 |
| (c) | **A** alcohol and heroin;<br><br>B antibiotics have no effect on mental health<br>C paracetamol has no effect on mental health<br>D antibiotics have no effect on mental health | | 1 |
| (d) | Any two from the following<br><br>brain/skin/sweat glands/muscles/blood vessels/arterioles;; | | 2 |
| | | | Total 10 |

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 2 of 11 · Question 2 part a.i · 1 mark

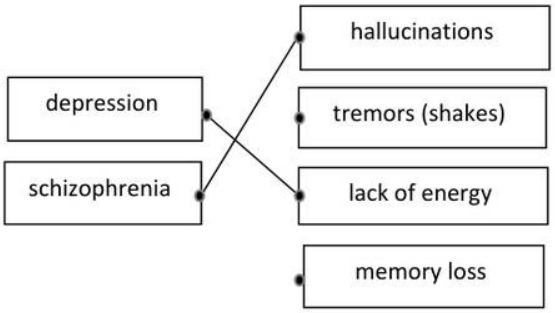
2022 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 3 of 11 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) | 2002; | | 1 |
| (ii) | 2005 (1.4) | | 1 |
| (iii) | 2.75:2.6;<br>1.1/1.06 :1; | | 2 |
| (iv) | low income/low self esteem/(domestic) violence/low social status/substance abuse; | | 1 |
| (b) |  | | 2 |
| (c) | **A** alcohol and heroin;<br><br>B antibiotics have no effect on mental health<br>C paracetamol has no effect on mental health<br>D antibiotics have no effect on mental health | | 1 |
| (d) | Any two from the following<br><br>brain/skin/sweat glands/muscles/blood vessels/arterioles;; | | 2 |
| | | | Total 10 |

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 3 of 11 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 4 of 11 · Question 8 part c.i · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 8 | (a) | (i) | triggers ovulation/causes release of the egg; | | 1 |
| | | (ii) | Three from:<br>• FSH stimulates growth/production of follicles in the ovary;<br>• more eggs mature/produced;<br>• increases the chances of more than one egg being released;<br>• each egg is fertilised by a sperm; | | 3 |
| | (b) | (i) | Three from:<br>• placenta supplies glucose/oxygen from the mother;<br>• twins have to share the supply of glucose/receive less glucose/oxygen;<br>• less aerobic respiration;<br>• less energy is released in the foetus and less growth occurs; | Allow nutrients for mp's 1 and 2 | 3 |
| | | (ii) | Two from<br>• same egg and sperm used;<br>• zygote splits/divides<br>• forming 2 embryos; | | 2 |
| | (c) | (i) | 75th | | 1 |
| | | (ii) | • baby is weighed regularly;<br>• growth should follow a percentile/graph/compare with (growth) chart/graph/compare mass to others of the same age;<br>• detect if the baby gains or loses too much mass; | | 3 |

Total 13 marks

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 4 of 11 · Question 8 part c.i · 1 mark

2023 JAN · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

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Type MS · Read or estimate a value from a graph, chart, table or image

Answer 5 of 11 · Question 4 part c.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 4 | (a) | any two from<br>• hearing/sound;<br>• touch;<br>• taste;<br>• smell; | | 2 |
| | (b) (i) | any two from<br>• as thickness decreases, focal length increases;<br>• interval between focal lengths increases as thickness decreases;<br>• (constant when two variables multiplied hence) inversely proportional; | | 2 |
| | (ii) | diameter decreases; | | 1 |
| | (iii) | any four from<br>• rays from near object diverging;<br>• ciliary muscle contracts;<br>• suspensory ligaments slacken/relax;<br>• so lens becomes thicker/more convex;<br>• more powerful lens/light refracted more/increased refraction;<br>• (allows it to focus because) focal length becomes shorter/decrease; | | 4 |
| | (c) (i) | between 575 and 580 nm; | | 1 |
| | (ii) | • 400 - 530;<br>• 130 nm; | ECF | 2 |
| | (iii) | • greater range of wavelengths can be seen/absorb light of different wavelengths;<br>• gives better colour perception/vision/differentiation/can see different colours; | | 2 |

Total 14

## Type MS · Read or estimate a value from a graph, chart, table or image

### Answer 5 of 11 · Question 4 part c.i · 1 mark

2023 JUN · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 6 of 11 · Question 4 part a.ii · 2 marks

2023 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

<table>
<tr>
<td>4 (a) (i)</td>
<td>
<table>
<tr><th>Activity</th><th>Pulse rate<br>in beats per minute</th></tr>
<tr><td>walking</td><td></td></tr>
<tr><td>jogging</td><td></td></tr>
<tr><td>running</td><td></td></tr>
</table>
</td>
<td>1 mark for valid table<br>1 mark for labels;<br>1 mark for units;</td>
<td>3</td>
</tr>
<tr>
<td>(ii)</td>
<td>
<table>
<tr><th>Activity</th><th>Pulse rate<br>in beats per minute</th></tr>
<tr><td>walking</td><td>94</td></tr>
<tr><td>jogging</td><td>116</td></tr>
<tr><td>running</td><td>133.5</td></tr>
</table>
</td>
<td>2 marks for all readings correct.<br>1 mark for 1 or 2 readings correct.<br><br>tolerance +/- ½ square</td>
<td>2</td>
</tr>
<tr>
<td>(iii)</td>
<td>Any four from five<br><br>• take resting pulse;<br>• method for measuring pulse rate e.g. finger on wrist/neck/electronic device;<br>• carry out an activity/exercise (for a set period of time/given time);<br>• measure pulse rate after activity;<br>• repeat (for each activity);</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>(b)</td>
<td>Any four from<br><br>• faster/more/greater muscle contraction during running;<br>• more energy/ATP required;<br>• faster rate of respiration;<br>• more oxygen/ glucose required/more $CO_2$ removed;<br>• increase blood flow/faster blood flow/more blood (to muscles);</td>
<td>ORA for all marking points</td>
<td>4</td>
</tr>
</table>

**Total for Question 4 = 13 marks**

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 7 of 11 · Question 3 part a.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | (i) | • time in minutes;<br>• 0-10, 11-20, 21-30, 31-40,41-50,51-60; | | 2 |
| | | (ii) | • 75 + 72 + 68; plus/minus 2 for each one<br>• 215 $cm^3$; | ECF | 2 |
| | | (iii) | • rapid reaction/a lot/large amount of gas produced, at start;<br>• rate slows/less gas (for next 10 minutes);<br>• drops a lot/little gas produced (after 40 minutes); | | 3 |
| | | (iv) | any three from<br>• at start plenty of substrate/many E-S complexes formed;<br>• substrate used up/less substrate available;<br>• reaction slows/less gas/product, produced/less enzyme action;<br>• most substrate used at end so little gas produced; | | 3 |
| | (b) | | • less gas produced/reaction lower/less;<br>• because active sites blocked; | | 2 |

Total 12

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 7 of 11 · Question 3 part a.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 8 of 11 · Question 2 part d.i · 1 mark

2020 JAN · Paper 01 · find part d.i in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 2 (a) | (i) | **The only correct answer is A**<br><br>*B is incorrect because it is the cervix*<br><br>*C is incorrect because it is the oviduct*<br><br>*D is incorrect because it is the uterus wall* | | 1 |
| | (ii) | **The only correct answer is D**<br><br>*A is incorrect because it is the vagina*<br><br>*B is incorrect because it is the cervix*<br><br>*C is incorrect because it is the oviduct* | | 1 |
| | (iii) | • oestrogen (1)<br>• lining (1)<br>• ovulation (1) | | 3 |
| (b) | | needed for the growth of bones (1) | | 1 |
| (c) | | ref antibodies (passed to baby)/contains the correct balance of nutrients/reduced risk of diabetes/allergies/obesity later in life/reference to bonding between mother and baby (1) | | 1 |
| (d) | (i) | more breastfed babies in 2010 compared with 1940 (1) | Allow reverse argument<br>A 78% vs 42% difference of 36% | 1 |
| | (ii) | • life expectancy shows a constant increase (1)<br>• despite number of babies breastfed increasing and decreasing (1) | | 2 |

**Total for Question 2 = 10 marks**

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 9 of 11 · Question 4 part a.i · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| **4(a)(i)** | ¼/25%/1:4/0.25; | | **(1)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **4(a)(ii)** | ½/50%/1:2/0.5/2<br>4; | | **(1)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **4(b)** | head decreases (in proportion);<br>legs increase (in proportion); | | **(2)** |

| Question Number | Answer | Additionalguidance | Mark |
|---|---|---|---|
| **4(c)** | • allowance for babies of different height;<br>• allows comparison to norm (in population);<br>• determine changes in proportion;<br>• over time / indicates/shows which parts of body are developing/growing; | | **Max (2)** |

**Total 6 Marks**

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 9 of 11 · Question 4 part a.i · 1 mark

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

**Paper 2: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 10 of 11 · Question 4 part a.ii · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| **4(a)(i)** | ¼/25%/1:4/0.25; | | **(1)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **4(a)(ii)** | ½/50%/1:2/0.5/2<br>4; | | **(1)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **4(b)** | head decreases (in proportion);<br>legs increase (in proportion); | | **(2)** |

| Question Number | Answer | Additionalguidance | Mark |
|---|---|---|---|
| **4(c)** | • allowance for babies of different height;<br>• allows comparison to norm (in population);<br>• determine changes in proportion;<br>• over time / indicates/shows which parts of body are developing/growing; | | **Max (2)** |

**Total 6 Marks**

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 10 of 11 · Question 4 part a.ii · 1 mark

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

**Paper 2: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

Type MS · Read or estimate a value from a graph, chart, table or image

Answer 11 of 11 · Question 3 part b · 1 mark

2020 JAN · Paper 02 · find part b in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | (i) | **The only correct answer is C** lens<br>***A** is incorrect because this is where the image is formed*<br>***B** is incorrect because that is what controls the size of the pupil*<br>***D** is incorrect because the pupil is the hole in the middle of the iris* | | 1 |
| | | (ii) | • retina(1)<br>• pupil (1)<br>• choroid(1) | | 3 |
| | (b) | | Value between 190 and 230 (μm) | | 1 |
| | (c) | (i) | • transmit (nerve/electrical) impulses(1)<br>• to the brain/CNS/relay neurones(1) | | 2 |
| | | (ii) | • less/no impulses able to reach the brain(1)<br>• vision blurred/less clear/less focussed/image not formed/seen/reference to blindness(1) | Allow signals | 2 |

**Total for Question 3 = 9 marks**